Band-1 Hadamard order 14424 – Classical Paley-I (q=14423); IN SPK via Paley-I; Gap 2052 Still OPEN

H_14424 Paley-I PASS; Paley-II FAIL/SKIP (q=7211 prime but ===3 mod 4 (needs ===1)); Kronecker FAIL/SKIP (4*3606; 3606===2 mod 4 NOT Hadamard-admissible; 3606=2*3*601); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-09

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. `2026-10-09_Band1_Hadamard_14420_PaleyI`; `2026-10-09_Band1_Hadamard_14412_PaleyI`; `2026-10-09_Band1_Hadamard_14408_PaleyI`; `2026-10-09_Band1_Hadamard_14388_PaleyI`; `2026-10-09_Band1_Hadamard_14356_PaleyII`; `2026-10-09_Band1_Hadamard_14348_PaleyI`; `2026-10-09_Band1_Hadamard_14328_PaleyI`; `2026-10-09_Band1_Hadamard_14324_PaleyI`; `2026-10-09_Band1_Hadamard_14304_PaleyI`; `2026-10-09_Band1_Hadamard_14252_PaleyI`; `2026-10-09_Band1_Hadamard_14244_PaleyI`; `2026-10-09_Band1_Hadamard_14208_PaleyI`; `2026-10-08_Band1_Hadamard_14160_PaleyI`; `2026-10-08_Band1_Hadamard_14144_PaleyI`; `2026-10-08_Band1_Hadamard_14108_PaleyI`; `2026-10-08_Band1_Hadamard_14088_PaleyI`; `2026-10-08_Band1_Hadamard_14084_PaleyI`; `2026-10-08_Band1_Hadamard_14072_PaleyI`; `2026-10-08_Band1_Hadamard_14052_PaleyI`; `2026-10-08_Band1_Hadamard_14012_PaleyI`; `2026-10-07_Band1_Hadamard_14000_PaleyI`; `2026-10-07_Band1_Hadamard_13968_PaleyI`; `2026-10-07_Band1_Hadamard_13964_PaleyI`; `2026-10-07_Band1_Hadamard_13932_PaleyI`; `2026-10-07_Band1_Hadamard_13908_PaleyI`; `2026-10-07_Band1_Hadamard_13904_PaleyI`; `2026-10-07_Band1_Hadamard_13884_PaleyI`; `2026-10-07_Band1_Hadamard_13880_PaleyI`; `2026-10-07_Band1_Hadamard_13860_PaleyI`; `2026-10-07_Band1_Hadamard_13832_PaleyI`; `2026-10-07_Band1_Hadamard_13808_PaleyI`; `2026-10-07_Band1_Hadamard_13800_PaleyI`; `2026-10-07_Band1_Hadamard_13764_PaleyI`; `2026-10-07_Band1_Hadamard_13760_PaleyI`; `2026-10-07_Band1_Hadamard_13752_PaleyI`; `2026-10-07_Band1_Hadamard_13724_PaleyI`; `2026-10-07_Band1_Hadamard_13712_PaleyI`; `2026-10-07_Band1_Hadamard_13692_PaleyI`; `2026-10-07_Band1_Hadamard_13688_PaleyI`; `2026-10-06_Band1_Hadamard_13680_PaleyI`; `2026-10-06_Band1_Hadamard_13628_PaleyI`; `2026-10-06_Band1_Hadamard_13620_PaleyI`; `2026-10-06_Band1_Hadamard_13592_PaleyI`; `2026-10-06_Band1_Hadamard_13568_PaleyI`; `2026-10-06_Band1_Hadamard_13524_PaleyI`; `2026-10-06_Band1_Hadamard_13500_PaleyI`; `2026-10-06_Band1_Hadamard_13488_PaleyI`; `2026-10-06_Band1_Hadamard_13464_PaleyI`;

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2026-10-05_Band1_Hadamard_13152_PaleyI; 2026-10-05_Band1_Hadamard_13148_PaleyI;
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2026-10-05_Band1_Hadamard_13044_PaleyI; 2026-10-05_Band1_Hadamard_13008_PaleyI;
2026-10-05_Band1_Hadamard_13004_PaleyI; 2026-10-05_Band1_Hadamard_12984_PaleyI;
2026-10-05_Band1_Hadamard_12980_PaleyI; 2026-10-05_Band1_Hadamard_12968_PaleyI;
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2026-10-03_Band1_Hadamard_12228_PaleyI; 2026-10-03_Band1_Hadamard_12212_PaleyI;
2026-10-03_Band1_Hadamard_12204_PaleyI; 2026-10-03_Band1_Hadamard_12164_PaleyI;
2026-10-03_Band1_Hadamard_12144_PaleyI; 2026-10-03_Band1_Hadamard_12120_PaleyI;
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2026-09-30_Band1_Hadamard_11484_PaleyI; 2026-09-30_Band1_Hadamard_11472_PaleyI;
2026-09-30_Band1_Hadamard_11468_PaleyI; 2026-09-30_Band1_Hadamard_11448_PaleyI;
2026-09-30_Band1_Hadamard_11444_PaleyI; 2026-09-30_Band1_Hadamard_11424_PaleyI;
2026-09-30_Band1_Hadamard_11412_PaleyI; 2026-09-30_Band1_Hadamard_11400_PaleyI;
2026-09-30_Band1_Hadamard_11384_PaleyI; 2026-09-30_Band1_Hadamard_11352_PaleyI;
2026-09-30_Band1_Hadamard_11312_PaleyI; 2026-09-30_Band1_Hadamard_11300_PaleyI;
2026-09-30_Band1_Hadamard_11288_PaleyI; 2026-09-30_Band1_Hadamard_11280_PaleyI;
2026-09-30_Band1_Hadamard_11252_PaleyI; 2026-09-29_Band1_Hadamard_11244_PaleyI;
2026-09-29_Band1_Hadamard_11240_PaleyI; 2026-09-29_Band1_Hadamard_11172_PaleyI;
2026-09-29_Band1_Hadamard_11160_PaleyI; 2026-09-29_Band1_Hadamard_11132_PaleyI;
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2026-09-29_Band1_Hadamard_10740_PaleyI; 2026-09-29_Band1_Hadamard_10724_PaleyI;
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2026-09-24_Band1_Hadamard_10332_PaleyI; 2026-09-24_Band1_Hadamard_10304_PaleyI;
2026-09-24_Band1_Hadamard_10272_PaleyI; 2026-09-23_Band1_Hadamard_10268_PaleyI;
2026-09-23_Band1_Hadamard_10224_PaleyI; 2026-09-23_Band1_Hadamard_10212_PaleyI;
2026-09-23_Band1_Hadamard_10164_PaleyI; 2026-09-23_Band1_Hadamard_10160_PaleyI;
2026-09-23_Band1_Hadamard_10152_PaleyI; 2026-09-23_Band1_Hadamard_10140_PaleyI;

2026-09-23_Band1_Hadamard_10112_PaleyI; 2026-09-23_Band1_Hadamard_10104_PaleyI; 2026-09-23_Band1_Hadamard_10100_PaleyI; 2026-09-23_Band1_Hadamard_10068_PaleyI.
Lab. /workspace/experiments/band1/hadamard_14424_paleyI/.
Cutoff. 2026-10-09T22:05:01Z (~6:05PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14424 only (verified max|HHˆT-nI|=0), via classical Paley-I q=14423 (prime ===3 mod 4). 14424 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14424: Paley-I **PASS** ($q = 14423$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=7211 prime but ===3 mod 4 (needs ===1)); Kronecker **FAIL/SKIP** (note $14424 = 4 \cdot 3606$; $3606 \equiv 2 \pmod 4$ NOT Hadamard-admissible; $3606 = 2 \cdot 3 \cdot 601$; do NOT invent H_3606). **14424** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture; 14424 is a long-known Hadamard order and this is a finite on-box verification, NOT a new-order claim. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14420+14412+14408+14388+14356+14348+14328+14324+14304+14252+14244+14208+14160+141** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **14424**.

- H_{14424} = Paley-I ($q = 14423$) – classical SPK
- Paley-II for 14424: **FAIL/SKIP** ($q = 7211$ prime but $\equiv 3 \pmod 4$ (needs $\equiv 1$))
- Kronecker for 14424: **FAIL/SKIP** ($4 \cdot 3606$; $3606 \equiv 2 \pmod 4$ NOT Hadamard-admissible; $3606 = 2 \cdot 3 \cdot 601$)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14424.py`

Results. overall_pass=True; units 748/748; row blocks 113/113 (block size 128); wall 2449.3855s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-09T22:05:01Z (~6:05PM ET)

Arithmetic

- Target $n = 14424 = q + 1$ with $q = 14423$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 7211$ prime but $\equiv 3 \pmod 4$ (needs $\equiv 1$) — FAIL/SKIP.

- Kronecker: $14424 = 4 \cdot 3606$; $3606 \equiv 2 \pmod 4$ NOT Hadamard-admissible; $3606 = 2 \cdot 3 \cdot 601$ — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14424 Paley-I	Paley-I q=14423	0	2412.1301
H_14424 Paley-II	FAIL/SKIP (q=7211 prime but ===3 mod 4 (needs ===1))	–	–
H_14424 Kronecker	FAIL/SKIP (4*3606; 3606===2 mod 4 NOT Hadamard-admissible; 3606=2*3*601)	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; …; **NOT** 8948/8644/8516/…) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
12488 in SPK	**True** (Paley-I q=14423)
Gap 2052	**OPEN**

Quantity	Value
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)
- 14424 is a long-known Hadamard order (Paley 1933 family); finite on-box verification only, NOT a new-order claim

Next optional

q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^2*71317 *NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2^2*59^2 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII))* | *q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2^2*31163 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII))* | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3^2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2^2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII))* | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*1389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2^2*3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII))* | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13^2*83 *NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7*167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2^2*3*7*167 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII))* | *q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5*941 NOT prime -> FAIL; Kronecker*

*14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2^2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3^2*1571 NOT prime -> FAIL; Kronecker 14140=4*3535; 3535=5*7101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2^2*5*7101 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7109->n=14220 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7109 prime ===1 mod 4; Paley-I q=14219=59*241 NOT prime -> FAIL; Kronecker 14220=4*3555; 3555=3^2*5*79 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555); 14220=2^2*3^2*5*79 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7129->n=14240 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7129 prime ===1 mod 4; Paley-I q=14239=3^2*1583 NOT prime -> FAIL; Kronecker 14240=4*3560; 3560=5*23*31 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3560); 14240=2^2*5*23*31 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7139->n=14260 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7139 prime ===1 mod 4; Paley-I q=14259=3*4753 NOT prime -> FAIL; Kronecker 14260=4*3565; 3565=5*23*31 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3565); 14260=2^2*5*23*31 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | FIRST Paley-II DIRECT after 14424 (and FIRST any direct route after 14424): q2=7213->n=14428 (q2=7213 prime ===1 mod 4; Paley-I q=14427=3^2*7*229 NOT prime -> FAIL; Kronecker 14428=4*3607; 3607 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3607); 14428=2^2*3607 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII); comes BEFORE the next Paley-I 14432) | next Paley-I DIRECT after 14424: q=14431->n=14432 (prime ===3 mod 4; no prime in 14424..14430; Paley-II q2=7215=3*5*13*37 NOT prime power (===3 mod 4) -> FAIL/SKIP; Kronecker 14432=4*3608; 3608=2^3*11*41 ===0 mod 4 Hadamard-admissible, H_3608 via Paley-I q=3607 prime ===3 mod 4 -> ALSO a classical route (note-only, NOT on box; Paley-I is the claim path); 14432=2?*11*41 ===0 mod 16) | Kronecker-only (note-only): none in 14428..14432 (3607 odd; 14432 has a direct Paley-I route) (or gap 2052 / Track B)